Lecture 10 (28/06/21)

Orthogonality

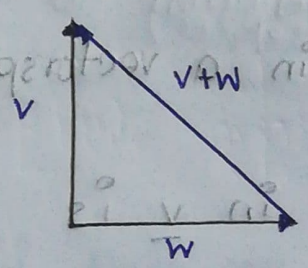
- Perpendicularity
- Two vectors are orthogonal if they are perpendicular

Two vectors are orthogonal if their dot product is 0.

$\therefore v = \begin{bmatrix} \\ \\ \end{bmatrix} \quad w = \begin{bmatrix} \\ \\ \end{bmatrix}$

$\boxed{\vec{v} \cdot \vec{w} = 0} \Rightarrow \boxed{v^T \cdot w = 0 \text{ or } v \cdot w^T = 0}$

Order doesn't matter (transpose)



$\|v\|^2 + \|w\|^2 = \|v + w\|^2$ [similar to pythagoras]

$\Rightarrow v \cdot v + w \cdot w = (v + w) \cdot (v + w)$

$\Rightarrow v^T \cdot v + w^T \cdot w = v^T \cdot v + v^T \cdot w + w^T \cdot w + w^T \cdot w$

$\Rightarrow v^T w + w^T \cdot v = 0$

$\Rightarrow 2 \cdot v^T \cdot w = 0$

$\Rightarrow v^T \cdot w = 0$

$\Rightarrow \boxed{\vec{v} \cdot \vec{w} = 0}$

Example:

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad y = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \quad x+y = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$$

$$\|x\|^2 = 14 \quad \|y\|^2 = 5 \quad \|x+y\|^2 = 19$$

[Individual dot products]

$$\therefore \|x\|^2 + \|y\|^2 = \|x+y\|^2$$

Orthogonality of Subspaces

Two subspaces being orthogonal to each other

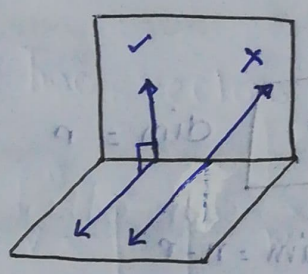
If we have two subspaces $\boxed{V}$ and $\boxed{W}$ in a vectorspace, these two are orthogonal if every vector in V is orthogonal to every vector in W

$v^T . w = 0$ for all vectors v in V and w in W.

Example:

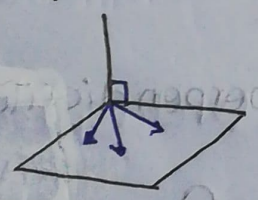
1. Floor and a wall ✗

Though it might seem they are perpendicular to each other, in reality, their vectors aren't always perpendicular.



2. Floor and corner of a room (intersection of two walls)

Always perpendicular



$\angle = 90°$

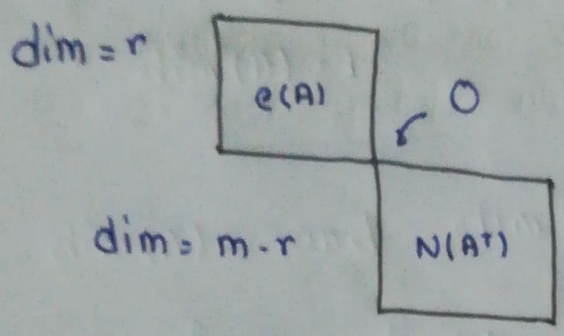
If there are two subspaces v and w in a vectorspace, orthogonality is possible only when,

$$\dim(v) + \dim(w) = \text{dim of vectorspace}$$

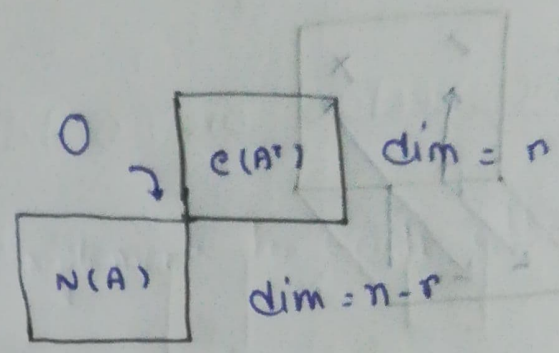
(impossible when $\dim(v) + \dim(w) > \dim(\text{vectorspace})$)

- 0 (zero) is the only point where the nullspace meets the rowspace.

Four fundamental subspaces.



$r + m - r = m$



$r + n - r = n$

Nullspace meets rowspace at 0 and they meet each other at 90°.

Every vector x in the nullspace is perpendicular to every row of A (rowspace) as $Ax = 0$

$N(A) \perp C(A^T)$

These vector subspaces aren't only orthogonal but also orthogonal complements. (complementing each other equal to the dimension of whole vector space.)

For $Ax = 0$,

$$\underset{A}{\begin{bmatrix} \leftarrow \text{row } 1 \rightarrow \\ \leftarrow \text{row } 2 \rightarrow \\ \leftarrow \text{row } 3 \rightarrow \\ \vdots \\ \leftarrow \text{row } m \rightarrow \end{bmatrix}} \underset{\substack{x \\ \text{(null space)}}}{\begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix}} = \underset{0}{\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}} \begin{matrix} \rightarrow (\text{row}1).x = 0 \\ \rightarrow (\text{row}2).x = 0 \\ \\ \rightarrow (\text{row } m).x = 0 \end{matrix}$$

Every row has a dot product value of 0 with x:

Every vector from nullspace is perpendicular to every combination of rowspace.

$\therefore$ The whole nullspace is orthogonal to the rowspace.

$$\boxed{C(A^T) \perp N(A) \text{ in } R^n}$$

$$\boxed{C(A) \perp N(A^T) \text{ in } R^m}$$

Example:

$$\underbrace{\begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{0}$$

$R^3 = \dim(A) + \dim(N(A))$

$= 1 + 2$

$r = 1$ $\dim(N(A)) = 2$

$\dim(A) = 1$

$\dim(A)$ and $\dim(N(A))$ are complement to each other

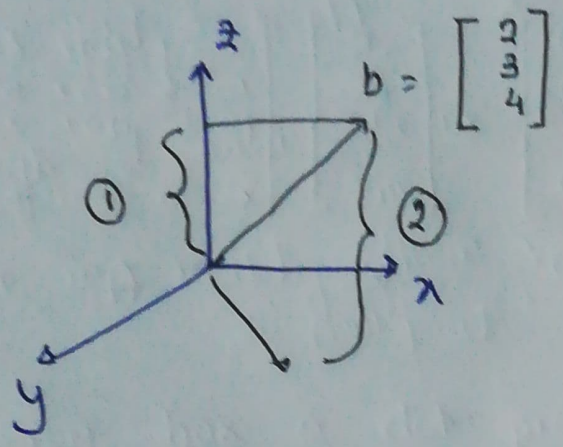
(no. of c^{ln}s of A)

Two orthogonal subspaces combinely fill the whole space

$\dim(V) + \dim(W) \neq \dim(\text{vectorspace})$

True for fundamental subspaces.

Projections



* What is the projection of b in z-axis?
* What is the projection of b in xy plane?
* What matrices produces these projections in the line (z-axis) and plane (xy plane)? And whats the relationship between them?

Ans:

Proj. of b in z-axis : $\begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$ matrix $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Proj of b in xy plane = $\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$